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Mode of action of milk and whey in the control of grapevine powdery mildew

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Abstract. Grapevine powdery mildew, caused by the fungus *Erysiphe (Uncinula) necator,* is a major disease affecting grape yield and quality worldwide. In conventional vineyards, the disease is controlled mainly by regular applications of sulfur and synthetic fungicides and, in organic agriculture, by sulfur and botanical and mineral oils. Research has identified milk and whey as potential replacements for synthetic fungicides and sulfur in the control of powdery mildew. Electron spin resonance and scanning electron microscopy were used to investigate the possible mode or modes of action of milk and whey in the control of powdery mildew. Electron spin resonance experiments showed that various components of milk produced oxygen radicals in natural light, which may have contributed to the reduction of severity of powdery mildew on treated leaves. Milk and whey caused the hyphae of *E. necator* to collapse and damaged conidia within 24 h of treatment. Hydrogen peroxide, applied as a source of free radicals, also caused collapse of the hyphae of *E. necator* but did not damage conidia, and appeared to stimulate germination. Lactoferrin (an antimicrobial component of milk) ruptured conidia, but damage to hyphae was not evident until 48 h after treatment. The results support the hypothesis that free radical production and the action of lactoferrin are associated with the control of powdery mildew by milk.

Additional keyword: organic viticulture.

Introduction

Vitis vinifera (European grape) cultivars are grown in many countries for wine, dried fruit and table grape production, but are most commonly used for wine production in Australia and Europe. Powdery mildew, caused by the fungus *Erysiphe (Uncinula) necator*, is a major disease of grapevine in many countries (Magarey *et al*. 1997) and causes losses due to damage to berries, reduced bunch weight (Chellemi and Marois 1992) and a decrease in wine quality (Ough and Berg 1979).

E. necator is controlled by regular applications of a number of materials, most commonly sulfur, as a wettable powder, dust or flowable formulation, oils, inorganic salts and/or synthetic fungicides. In organic viticulture, sulfur and vegetable oil formulations are mainly used to control powdery mildew but other materials such as foliar coatings, e.g. clay and silicates, and inorganic salts are used by some growers.

Research has identified potential replacements for synthetic fungicides and sulfur in the control of powdery mildew, including milk and whey (Crisp *et al*. 2002). Weekly applications of milk at concentrations of up to 50% proved more effective than weekly applications of fenarimol and benomyl in the control of powdery mildew of zucchini (caused by *Sphaerotheca fuliginea*) (Bettiol 1999). Milk, applied at rates between 1 : 5 and 1 : 10 (v/v), and whey powder at 25 g/L, provided effective control of grapevine powdery mildew in greenhouse and field trials (Crisp *et al*. 2002). Milk and whey appear to act as contact fungicides against the superficial powdery mildew fungus.

There have been a number of explanations for the action of milk, including the anti-fungal action of the fatty acids, the production of free radicals when exposed to UV light or the creation of osmotic imbalance due to salts and other components. Bettiol (1999) suggested that milk may have a direct effect on the fungus or may induce systemic resistance to powdery mildew in zucchini. There is also evidence that exposure of milk to the ultraviolet radiation in sunlight results in the photogeneration of superoxide anions (Korycha-Dahl and Richardson 1978) and oxygen radicals that interfere with the cell membranes of *Phytophthora infestans* (Jordan *et al*. 1992). The production of free radicals when methionine and riboflavin have been exposed to UV

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light has been shown to control powdery mildew (Tzeng and DeVay 1989). Free radical production can be measured using electron spin resonance spectrometry (ESR) (Tzeng and De Vay 1984, 1989).

Components of milk and whey, particularly lactoferrin and lactoperoxidase, have been extensively researched as antimicrobial agents for use in human medicine and food preservation (Batish *et al*. 1988; Modler *et al*. 1998; Kuipers *et al*. 1999; Sallmann *et al*. 1999; Kanyshkova *et al*. 2000; Samaranayake *et al*. 2001). Lactoferrin, an 80 kDa iron-binding glycoprotein, binds to the membranes of various bacteria and fungi, causing damage to membranes and loss of cytoplasmic fluids (Batish *et al*. 1988; Kuipers *et al*. 1999; Sallmann *et al*. 1999; Samaranayake *et al*. 2001). However, research has focused largely on Gram-negative bacteria and other species primarily involved in food spoilage, and on fungi related to human health, especially *Candida* spp. (Kuipers *et al*. 1999; Kanyshkova *et al*. 2000; Samaranayake *et al*. 2001). The optimal concentration of lactoferrin for control of *Escherichia coli* and *Salmonella typhi* is 0.2 g/L (Batish *et al*. 1988). The concentration of lactoferrin ranges from 20 to 200 mg/L in bovine milk and from 56 to 164 mg/L in whey (Riechel *et al*. 1998).

Lactoperoxidase, a known antimicrobial protein, is present in milk at ~30 mg/L (Modler *et al*. 1998). The ability of lactoperoxidase to control *E. coli, Lactococcus lactis* and some other bacteria varied with the concentration of hydrogen peroxide and thiocyanate, and was reduced in the presence of low or excessive concentrations of hydrogen peroxide. Whey contains 5.6–28.6 mg/L hydrogen peroxide (Modler *et al*. 1998). Research into the ability of lactoperoxidase to control fungi appears limited.

Establishing the mode of action of milk and whey will provide information that will assist the optimisation of spray timing and intervals in the vineyard. The identification of the active fractions of milk and whey may also enable the development of additional commercial controls of powdery mildew that are suitable for use in organic and conventional vineyards. Thus, ESR and scanning electron microscopy (SEM) were used to investigate the mode of action of milk on grapevine powdery mildew.

Methods

Detection of free radical activity

Milk, whey and selected milk components, including lactoferrin and apo-lactoferrin (iron depleted), were assessed for the presence and production of active oxygen radicals using ESR. A Varian E-12 EPR spectrometer, operating at X-band (~9.1 GHz) at the Department of Physics at Monash University, Melbourne, Victoria, was used. All measurements were performed at room temperature (~22°C). The specimens were placed in standard, special quartz EPR tubes, with an internal diameter of ~2 mm. The sensitivity of the apparatus was confirmed using the standard 'weak pitch' sample supplied by Varian, which gives a stable signal for carbon radicals in pitch that is easily calibrated against other known standards. All dry samples were tested undiluted for free radical signals. The apparatus was not sufficiently sensitive to observe free radical signals at the dilutions used in the field.

Scanning electron microscopy

Infected leaf tissue treated with various test materials was examined in a Philips XL30 field emission SEM at Adelaide Microscopy. The microscope was equipped with an Oxford Instruments CT1500 HF Cryo. The cryogenic technique, involving minimal specimen preparation, was selected to reduce the likelihood of damage to hyphae and conidia during preparation.

The following treatments were examined as aqueous solutions: milk (1 : 5 and 1 : 10 dilutions of full cream pasteurised bovine milk); whey (30 g/L whey powder); whey protein (30 g/L); lactoferrin (from bovine colostrum, 20 mg/L, Sigma Chemicals); lactoperoxidase (from bovine colostrum, 10 g/L, Sigma Chemicals); hydrogen peroxide (1 mL/L); Synertrol Horti-Oil (2 mL/L, Organic Crop Protectants Pty Ltd) and Synertrol Horti-Oil (2 mL/L) mixed with potassium bicarbonate (4 g/L Ecocarb, Organic Crop Protectants Pty Ltd). These application rates were equivalent to rates that were used in previous greenhouse or field experiments (Crisp *et al*. 2002).

Young, near-fully expanded leaves with no noticeable powdery mildew were excised from *V. vinifera* cv. Viognier or cv. Cabernet Sauvignon that had been grown in the greenhouse. The leaves were surface sterilised by soaking in a solution of 0.5 g/L sodium hypochlorite and Tween 80 for 3 min. The leaves were then rinsed in sterile distilled water three times and allowed to dry on sterile paper towels before being placed in tissue culture dishes (Falcon), 100 mm in diameter and 20 mm deep, containing 1.5% agar (Bitek, Difco Laboratories, Michigan, USA). Leaves were placed adaxial side uppermost on four sterile toothpicks with the petiole embedded in the agar (Evans *et al*. 1996). Spores of *E. necator,* collected from vines maintained in tissue culture using a modified cyclone separator, were brushed onto the leaves using a sterile artist's brush (Evans *et al*. 1996). The leaves were then incubated in a growth cabinet for 2 weeks at 25°C with a 12 h light/dark cycle to allow the fungus to grow on the leaf surface and form conidia. Leaves on which no powdery mildew developed were discarded.

Test materials were applied using an Atomizer reagent sprayer (Alltech Associates Inc., Belgium) to optimise leaf coverage. In some experiments, half of the leaf surface was protected by a plastic shield to examine treatment effects on the same leaf and to explore the possibility of systemic activity. The dishes were left open in a laminar flow cabinet to allow the sprayed leaves to dry; the lids were then replaced and the dishes were stored at ~15/25°C night/day, in natural light. Additional plates containing leaves treated with milk, whey, lactoperoxidase or lactoferrin were incubated in darkness by wrapping the plates in foil. Leaf segments, ~10 × 5 mm, were cut from the treated leaves and frozen in semi-solid nitrogen, then transferred under vacuum to the preparation chamber. In the case of leaves that were partly shielded, the leaf segment was cut to represent both treated and untreated leaf surfaces. The sample was coated with platinum, then placed on the microscope stage (held at <–150°C) and examined. Leaves were examined 24 h after treatment, unless otherwise stated.

Leaves treated with milk 30 min, 6, 12, 18 or 48 h prior to examination were also observed to assess response over time. Additional leaves treated with whey, whey protein and lactoferrin were incubated in natural light for 48 h prior to examination. Infected leaves, untreated or sprayed with sterile, reverse osmosis water 24 h prior to examination, were used as controls.

Observations of the hyphae and conidia of *E. necator* on untreated leaves and on leaves sprayed with sterile, reverse osmosis water only were used as a reference when observing fungal structures on leaves sprayed with the test materials. The percentage of damaged hyphae was estimated based on the relative proportion of hyphae that appeared undamaged and similar in appearance to hyphae on untreated and water-sprayed leaves. Estimates of damage to the conidia were based on counts of conidia that appeared damaged and those that appeared similar to conidia on untreated and water-treated leaves.

A second series of experiments was conducted using leaves with older colonies of *E. necator* bearing cleistothecia, which were collected from the greenhouse, placed directly into tissue culture dishes, as above, and treated 24 h prior to examination. Untreated leaves were used as a reference for the appearance of natural deterioration of hyphae and conidia, to allow comparison with those structures following treatment with test materials.

Results

Detection of free radical activity

EPR spectrometer signals and outputs for mixtures are qualitative rather than quantitative and, while showing the presence or absence of free radical species, do not provide accurate information on their relative abundance, as they do for a single substance. All samples tested in the EPR spectrometer showed a symmetrical free radical signal with no structure and a linewidth of ~4 gauss. The control, sterile, reverse osmosis water had no free radical signal. Both lactoferrin and apo-lactoferrin produced a signal with the same shape and structure and of a similar signal to noise ratio; with the saturation behaviour of the signal, this suggests that the signal is at least partially independent of the iron interaction of the molecule.

Scanning electron microscopy

Approximately 90% of *E. necator* hyphae from 2-week-old colonies on untreated and water-treated leaf segments appeared turgid and undamaged, with the remaining 10% either collapsed or damaged (Fig. 1). Likewise, more than 90% of conidia appeared undamaged 24 h after spraying with water. There was no obvious difference in the appearance of conidia and hyphae on samples collected from control leaves incubated in natural light or where light was excluded. Conidia produced germ tubes at one end and close to the leaf surface. Colonies that were more than 6 weeks old displayed large areas of collapsed hyphae, and conidia were often collapsed and cracked (Fig. 1*b*). Cleistothecia on untreated leaves were roughly spherical in shape and appeared turgid.

Milk (1 : 10 dilution) or whey (25 g/L), applied to leaves incubated in natural light 24 h prior to examination, resulted in extensive damage to both hyphae and conidia but not to cleistothecia. Most (80–90%) of the hyphae were collapsed and, in some cases, appeared to have ruptured (Fig. 2*a*). A similar proportion of conidia were collapsed, cracked or ruptured with extrusion of cell contents along the cracks (Fig. 2*b*). However, undamaged conidia were able to germinate. No symptoms were noted on fungal structures on untreated areas of the same leaves. For treated leaves kept in the dark for the 24 h prior to examination, appearance of the conidia was similar to that on leaf segments incubated in light, but only 20% of the hyphae were damaged. Cleistothecia appeared undamaged and similar in appearance to those on untreated leaves.

The extent of damage to fungal structures on leaf samples observed 30 min after spraying with milk was difficult to assess accurately as the samples were not completely dry and milk droplets obscured much of the leaf surface. The cytoplasmic extrusions were present on ~20% of conidia observed 6 h after treatment. Up to 80% of the hyphae observed were damaged at this time; however, many were only partially, rather than completely, collapsed as was the case with samples treated 12, 24 or 48 h previously. There were no obvious differences in the extent of damage to conidia and hyphae of *E. necator* on samples from leaves sprayed with milk and observed 12, 24 or 48 h after treatment.

Application of whey protein to the leaves exposed to natural light and maintained at 20–25°C for 24 h prior to examination resulted in damage to fungal structures similar to that observed on leaves that had been sprayed with milk and incubated in darkness. The majority of hyphae (>80%) were turgid and similar in appearance to those on untreated leaves. Conidia displayed cracking and rupturing. Cleistothecia appeared slightly collapsed and dehydrated (Fig. 3). Examination of samples 48 h after application of whey protein showed complete collapse of conidia and collapse of up to 50% of the hyphae.

When leaves were sprayed with lactoferrin 24 h prior to assessment and incubated in natural light at 20–25°C, ~70% of the hyphae of *E. necator* appeared undamaged. The remaining 30% had lost turgidity and, in some places, appeared to have ruptured. Approximately 90% of the conidia were cracked and ruptured, with leakage of cell contents. Leaves sprayed with lactoferrin 48 h prior to assessment and incubated similarly, showed collapse of ~50% of hyphae and all conidia (Fig. 4).

Up to 20% of conidia had split on leaves 24 h after treatment with lactoperoxidase and incubation in natural light at 20–25°C, but the remaining 80% appeared undamaged and, in some cases, to have germinated normally. Approximately 70% of hyphae appeared turgid and undamaged, whereas the remaining 30% had lost turgidity and collapsed.

Hyphae were collapsed and ~50% of conidia appeared disfigured where mixtures of Synertrol Horti-Oil and Ecocarb were applied to infected leaves. The integrity of hyphae was maintained only in areas that were not treated. The conidia did not show cytoplasmic extrusions, but germ tubes had collapsed. Residues were more noticeable on leaves sprayed with Synertrol Horti-Oil plus Ecocarb than on leaves sprayed with other materials in the experiments. The damage to hyphae and conidia seen on leaves sprayed with Synertrol Horti-Oil plus Ecocarb was not evident on leaves sprayed with Synertrol Horti-Oil alone.

Approximately 80% of the conidia treated with hydrogen peroxide 24 h prior to SEM examination appeared undamaged, but the hyphae had collapsed and ruptured

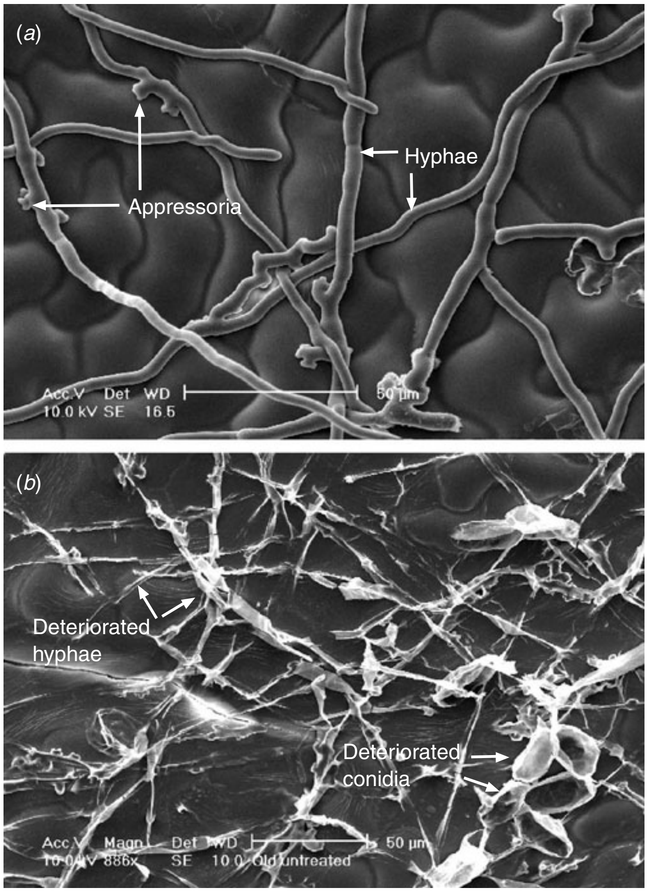

Fig. 1. Vine leaf surface with *Erysiphe necator* sprayed with water. *E. necator* on the upper surface of a detached leaf of *Vitis vinifera* cv. Viognier, sprayed with sterile reverse osmosis water 24 h prior to examination and incubated in natural light at 20–25°C. (*a*) 15-day colony and (*b*) 42-day colony.

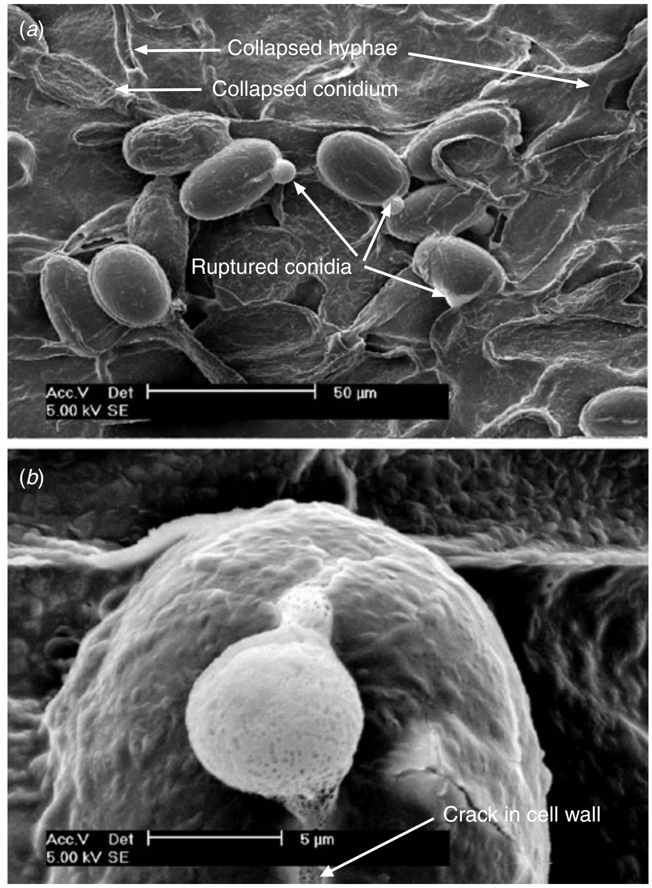

Fig. 2. Vine leaf surface with 15-day colony of *Erysiphe necator*, 24 h after milk treatment. *E. necator* on the upper surface of detached leaf of *Vitis vinifera* cv. Viognier, sprayed with a 1 : 10 dilution of milk and exposed to natural light at 20–25°C after treatment. (*a*) Collapsed hyphae and conidia and ruptured conidia are indicated by arrows. (*b*) Close up of ruptured conidium.

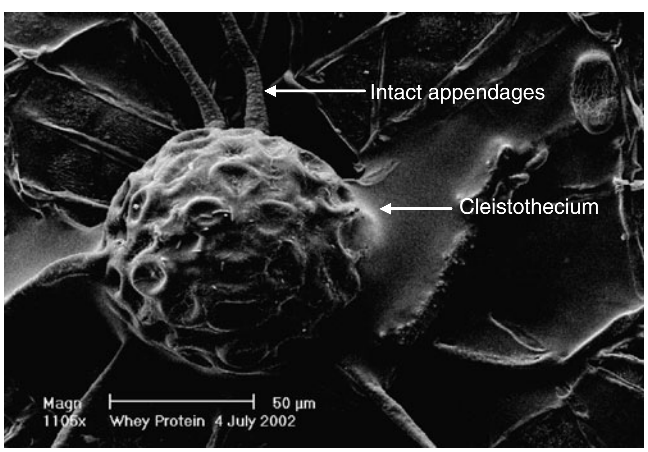

Fig. 3. Vine leaf surface with cleistothecium of *Erysiphe necator*, 24 h after whey treatment. Cleistothecium on the upper surface of detached leaf of *Vitis vinifera* cv. Viognier collected from the greenhouse, sprayed with 30 g/L whey protein. After treatment, leaf was incubated in natural light at 20–25°C.

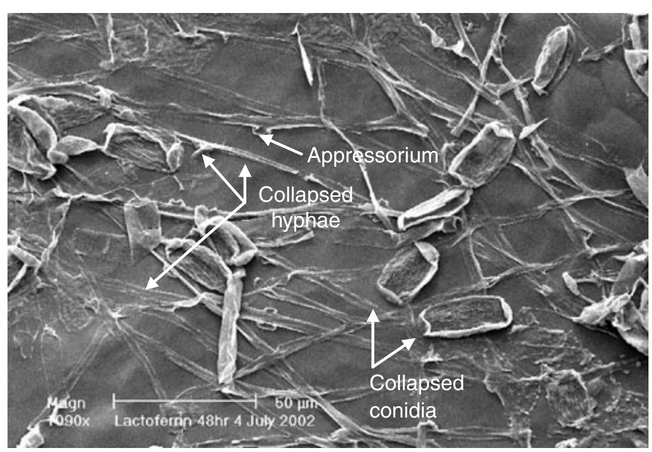

Fig. 4. Vine leaf surface with 15-day colony of *Erysiphe necator*, 48 h after lactoferrin treatment. Conidia of *E. necator* on the upper surface of detached leaf of *Vitis vinifera* cv. Viognier, sprayed with 20 mg/L lactoferrin. After treatment, leaf was incubated in natural light at 20–25°C.

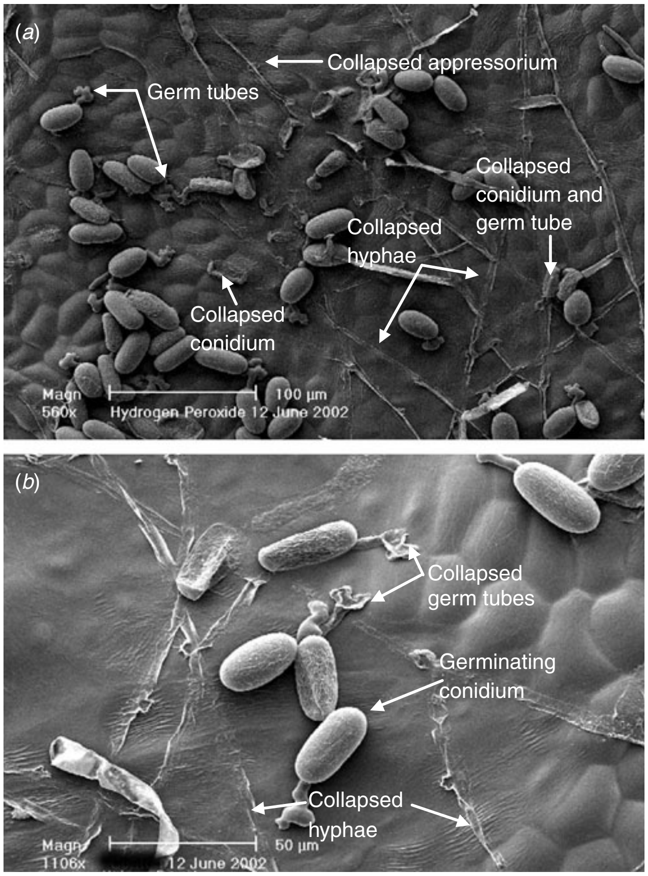

Fig. 5. Vine leaf surface with 15-day colony of *Erysiphe necator*, 24 and 48 h after hydrogen peroxide treatment. Hyphae and conidia of *E. necator* on the upper surface of detached leaf of *Vitis vinifera* cv. Viognier, sprayed with 0.1% hydrogen peroxide solution, showing collapsed hyphae and germinating conidia. After treatment, leaf was incubated in natural light at 20–25°C. (*a*) 24 h after treatment and (*b*) 48 h after treatment.

(Fig. 5*a*). The remaining conidia had collapsed completely. Over 50% of the intact conidia on the hydrogen peroxide treated leaf segments appeared to have germinated (Fig. 5*a*), compared with ~10% of conidia on untreated leaf segments. Cytoplasmic extrusions were visible on some partially collapsed hyphae on leaf segments treated with hydrogen peroxide. Approximately 50% of the germ tubes that had emerged from conidia appeared to have collapsed when examined 48 h after the application of the hydrogen peroxide solution to leaves that were incubated in natural light (Fig. 5*b*).

Discussion

Several of the materials tested caused conidia and hyphae of *E. necator* to collapse after incubation for 24 h in natural light. Milk, whey and whey protein damaged both hyphae and conidia, whereas damage 24 h after treatment appeared to be restricted to conidia or hyphae in the case of lactoferrin and hydrogen peroxide, respectively. However, both hyphae and conidia were affected 48 h after treatment with these materials. When infected leaves were incubated in the dark after treatment with a 1 : 10 dilution of milk, the damage to conidia was unchanged but the damage to hyphae was greatly reduced compared with those incubated in natural light.

Free radicals were produced by all samples when exposed to natural light. Free radicals have been shown to reduce the severity of powdery mildew and may have contributed to the reduction in powdery mildew severity observed in greenhouse and field experiments. Tzeng and DeVay (1989) found that exposure of methionine, riboflavin and sulfur-rich amino acids to light resulted in the production of free radicals that was associated with reduced severity of powdery mildew on grapevines. In the field, fluctuations in the intensity of solar UV radiation may influence the production of free radicals from the milk or whey solids. Should it occur, this variation in free radical production might have implications for the timing of spray applications in the field. Spraying on bright, warm days could result in greater free radical production than application on cooler, cloudy days. This may also have implications for the efficacy of the test materials in cool climates where the intensity of natural light is lower, leading to a need to increase concentrations of materials or reduce spray intervals.

Hydrogen peroxide, which is known to cause DNA strand breaks and membrane disruption (Hoffman and Meneghini 1979; Imlay and Lin 1988) at a concentration similar to that in milk, damaged hyphae and appeared to stimulate germination of conidia that were not ruptured. However, examination of powdery mildew-affected leaf segments 48 h after application revealed that ~50% of the germ tubes emerging from such conidia had collapsed. Whether this was a direct effect of the hydrogen peroxide, failure of the germ tubes to form penetration pegs or appressoria, the induction of a plant defence response or some other effect is not known.

SEM observations of powdery mildew on leaves treated with a 1 : 5 or 1 : 10 dilution of milk and incubated in natural light for 6–48 h revealed damage to both hyphae and conidia that was not evident on leaves sprayed with water or left untreated. The damage to hyphae of *E. necator* sprayed with a 1 : 10 dilution of milk or 0.1% hydrogen peroxide was similar, whereas only the conidia treated with milk appeared damaged, suggesting that milk had fungicidal properties. This supports the results from previous greenhouse trials, in which milk and whey appeared to provide eradicative control of grapevine powdery mildew (Crisp *et al*. 2000).

When light was excluded after treating powdery mildew-affected leaves with a 1 : 10 dilution of milk, the damage to hyphae was considerably less extensive than when leaves were incubated in natural light; conidia were ruptured regardless of the presence of light. The reduced damage to hyphae in the absence of light suggests that the fractions of milk active against hyphae and conidia in the 24 h after application are different or that they act in a different manner. Riboflavin–methionine mixtures reduced the severity of disease due to *P. infestans* only when exposed to light, which generated hydroxyl radicals or superoxide anions depending on pH (Jordan *et al*. 1992). If the damage to hyphae was caused by induction of resistance in the plant due to the application of milk and whey, exclusion of light should not reduce the efficacy of the treatments.

The rupturing of conidia of *E. necator* was observed on leaves sprayed with milk, whether incubated in natural light or in darkness, and on leaves treated with whey, whey protein and lactoferrin. If lactoferrin bound to membranes of *E. necator* in these experiments and altered their permeability, possibly disrupting the osmotic balance of the cells (O. Schmidt, pers. comm.), increased internal pressure might cause such rupture. Further research is required to establish if lactoferrin binds to the cell membranes of *E. necator* and is responsible for this damage. However, the visually identical nature of the damage caused to conidia by lactoferrin, milk and whey suggests that lactoferrin is an active component of milk and whey in these reactions. Although lactoferrin caused more collapse of *E. necator* hyphae at 48 h than at 24 h after exposure, the majority of the damage caused to hyphae after treatment with milk, whey or hydrogen peroxide occurred within 24 h of application. Possibly, components of milk other than lactoferrin also caused damage to the hyphae, or the interaction with other components of milk increased the activity of lactoferrin against hyphae (Crisp *et al*. 2003).

Application of lactoperoxidase, while resulting in damage to ~20% of the conidia and 30% of the hyphae of *E. necator* on sprayed leaf segments, caused less damage than did milk or whey. However, the ability of lactoperoxidase to control *E. coli, L. lactis* and other bacteria varied with the concentration of hydrogen peroxide and thiocyanate added (Modler *et al*. 1998). Damage may have increased if suitable amounts of thiocyanate or hydrogen peroxide had been included with lactoperoxidase. The extent of damage to *E. necator* observed in the experiments reported here suggests that lactoperoxidase is a component of milk and whey which is active in the control of powdery mildew and, as thiocyanate and hydrogen peroxide are present in milk and whey, the effects of milk and whey on powdery mildew may be greater than when lactoperoxidase is applied alone. Further *in vitro* and greenhouse experiments are needed to determine the effect of the additives if lactoperoxidase is to be used in the control of powdery mildew.

Although some active components of milk and whey, such as lactoferrin, have been identified, further evaluation of their mode of action is required. A thorough understanding of the mechanisms by which components such as lactoferrin interact with *E. necator* to cause physical damage will assist in calculating spray timing and concentrations to achieve maximum cost-effective disease control.

Investigation of other components of milk is required to assess their ability to control powdery mildew alone and in combination. A thorough understanding of the mechanisms may assist in the development of new products for the control of powdery mildew and other plant diseases. For example, if a second protein or molecule is involved in the interaction of *E. necator* and lactoferrin, mixtures of the two may be more effective than either compound used in isolation. As other components of milk that contribute to the control of powdery mildew are identified, their modes of action should be investigated.

Milk, whey, lactoferrin, and mixtures of botanical oil plus bicarbonate have potential as alternatives to sulfur and synthetic fungicides for the control of powdery mildew in grapevines. Spray programs involving milk or whey and oil plus bicarbonate mixtures would reduce concerns about the perceived negative impacts of relying on applications of oil plus bicarbonate or sulfur all season.

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